Unit 10: Transport layer-protocols

CONTENTS

Objectives

- understanding services of transport layer
- difference between connection-oriented and connection-less service.
- Reliable vs Unreliable Delivery
- TCP 3-Way Handshake Process

Introduction

The fourth layer from the top is the transport layer. The transport layer's primary function is to provide direct connectivity services to application processes operating on various hosts. The transport layer allows application processes operating on separate hosts to communicate logically. About the fact that application processes on separate hosts are not physically connected, application processes use the transport layer's logical connectivity to transmit messages to one another. End systems are equipped with transport layer protocols, but network routers are not. The network programmes on a computer network may use more than one protocol. TCP and UDP, for example, are two transport layer protocols that supply the network layer with separate networks. Multiplexing/demultiplexing is supported by all transport layer protocols. It also offers other features such as secure data transmission, guaranteed bandwidth, and guaranteed latency. Each programme in the application layer is capable of sending a message over TCP or UDP as seen in Figure 1. These two protocols are used by the programme to communicate. In the internet layer, both TCP and UDP can connect with the internet protocol. The programmes have access to the transport layer and can read and write to it. As a result, contact can be described as a two-way mechanism.

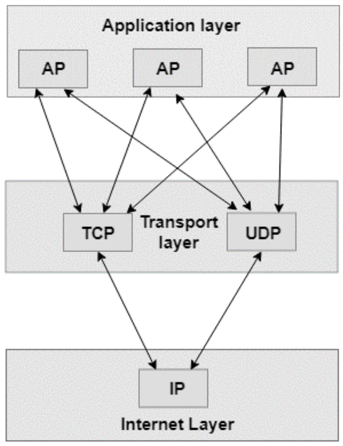

Figure 1 Transport Layer

10.1 Services provided by the Transport Layer

The transport layer provides facilities that are identical to those offered by the data link layer. The data link layer offers services within a single network, while the transport layer provides services through several networks in an internetwork. The physical layer is controlled by the data link layer, while the lower layers are controlled by the transport layer.

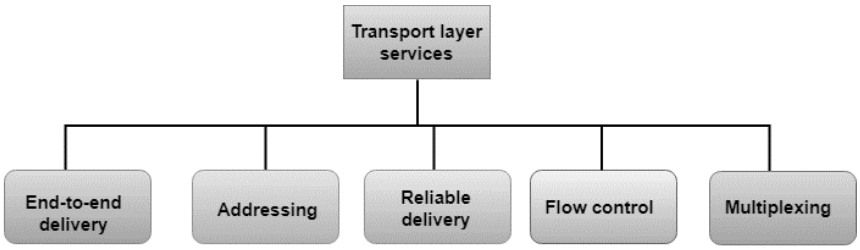

Figure 2 Services Provided by Transport Layer

The five types of services offered by transport layer protocols are as follows:

- **End-to-end delivery**
- **Addressing**
- **Reliable delivery**
- **Flow control**
- **Multiplexing**

End to End delivery

The transport layer sends the entire message to its intended recipient. As a result, it means that a whole message is sent from source to destination.

Reliable delivery

Through retransmitting missing and corrupted packets, the transport layer delivers redundancy services. The reliable delivery has four aspects:

- ***Error control***
- ***Sequence control***
- ***Loss control***
- ***Duplication control***

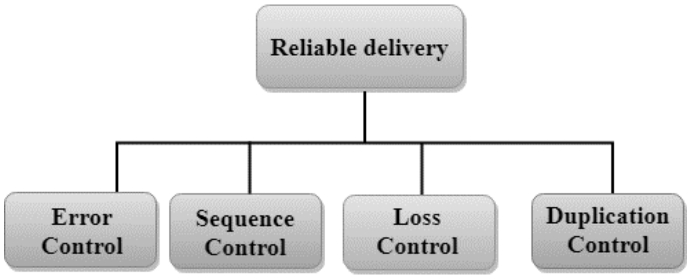

Figure 3 Reliable Delivery

Error Control

- The primary role of reliability is **Error Control**. In reality, no transmission will be 100 percent error-free delivery. Therefore, transport layer protocols are designed to provide error-free transmission.
- The data link layer also has an error management feature, but it only guarantees error-free transmission from node to node. End-to-end reliability is not guaranteed by node-to-node reliability.
- The data link layer examines each network for errors. If an error occurs within one of the routers, the data link layer would not be able to detect it. It only detects errors that occurred between the beginning and the end of the connection. As a result, the transport layer checks for errors from beginning to end to guarantee that the packet arrives intact.

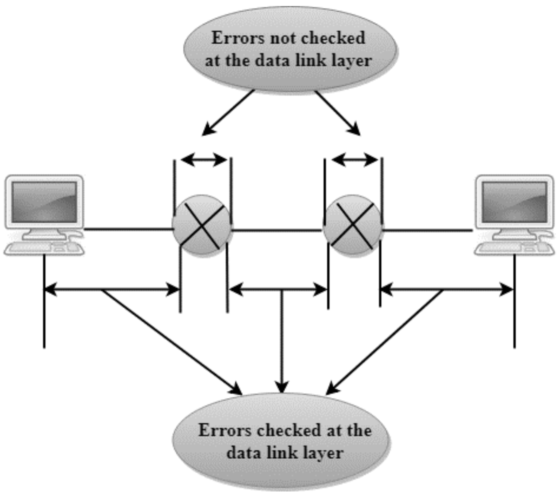

Figure 4 Error at Data Link Layer

Sequence Control

Sequence monitoring, which is applied at the transport layer, is the second component of reliability.

The transport layer is in charge of ensuring that packets obtained from the upper layers can be used by the lower layers on the receiving end. It means that the different segments of a transmission can be properly reassembled on the receiving end.

Loss Control

The third factor of durability is loss control. The transport layer means that all of a transmission's fragments, not just any of them, arrive at their destination. A transport layer assigns sequence numbers to all transmission fragments on the sending end. The receiver's transport layer will use these sequence numbers to locate the missing portion.

Duplicate Control

The fourth factor of durability is duplication control. The transport layer ensures that no redundant data reaches its intended location. Sequence numbers are used to locate missing packets, as well as to identify and discard duplicate fragments by the recipient.

Flow Control

Flow management is used to keep the sender from sending too many data to the recipient. When a receiver is overwhelmed with files, it discards packets and requests that they be retransmitted. As a result, network interference increases, lowering system performance. Flow regulation is handled by the transport layer.

Multiplexing

Multiplexing is used by the transport layer to increase transmission reliability.

- *Multiplexing can occur in two ways:*
    - ***Upward multiplexing*:** Upward multiplexing means multiple transport layer connections use the same network connection. To make more cost-effective, the transport layer sends several transmissions bound for the same destination along the same path; this is achieved through upward multiplexing as seen in .

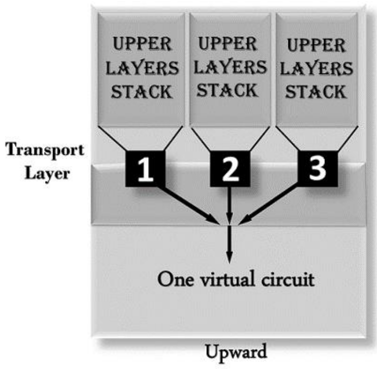

Figure 5 Upward Multiplexing

Downward multiplexing refers to the use of multiple network links through one transport layer link. Downward multiplexing requires the transport layer to divide a link into multiple directions in order to increase throughput. When networks have a low or sluggish bandwidth, this form of multiplexing is used as seen in Figure 6.

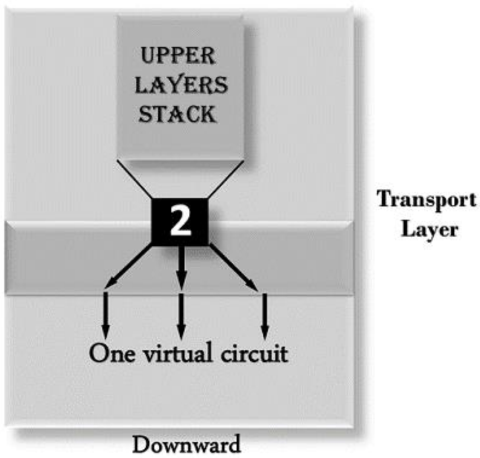

Figure 6 Downward Multiplexing

10.2 Difference between Connection-Oriented and Connectionless Service

To build communications between two or more computers, Connection-oriented and Connection-less Services are used. The establishment and termination of a link for sending data between two or more devices is part of a connection-oriented service. Connectionless operation, on the other hand, does not necessitate the establishment of any connection or termination mechanism in order to transmit data over a network.

Connection-oriented service

A network infrastructure that was planned and built since the telephone system is known as a connection-oriented service. Until sending data over the same or separate networks, a link-oriented service is used to establish an end-to-end connection between the sender and the receiver. Packets are delivered to the recipient in the same order as they were sent to the sender in connection-oriented operation. It employs a handshake procedure to establish a link between the recipient and the sender in order to transfer data over the network. As a result, it is often referred to as a dependable network infrastructure. Assume a sender wishes to transmit data to a recipient. The sender then sends a request packet in the form of a SYN packet to the recipient. After that, the receiver sends a (SYN-ACK) signal/packets in response to the sender's message.

That signifies the recipient's acceptance of the sender's request to begin contact with the receiver. The letter or data will now be sent to the recipient by the sender.

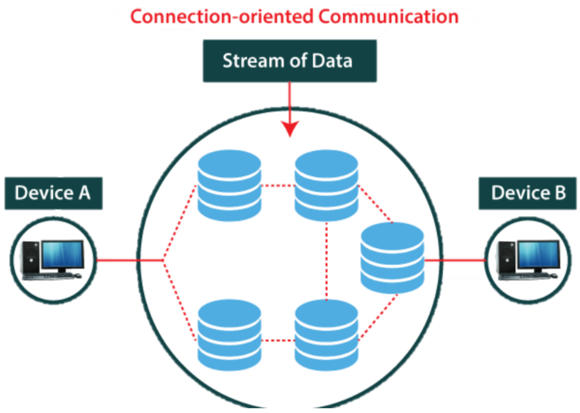

Figure 7 Connection-Oriented Communication

Similarly, a recipient may reply or send data in the form of packets to the sender. A sender may terminate a link by sending a signal to the receiver after successfully exchanging or transmitting data. As a result, we can conclude that it is a dependable network operation.

Example of connection-oriented is TCP.

TCP

TCP (Transmission Control Protocol) is a connection-oriented protocol that establishes links in the same or separate networks to facilitate communication between two or more computer devices. That is the most important protocol for transferring data from one end to the other using the internet protocol. TCP/IP is the abbreviation for Transmission Control Protocol/Internet Protocol.

10.3 Connectionless Service

A link is similar to the postal system in that each letter follows a separate path from the source to the destination address. In a network system, a connectionless service is used to transmit data from one end to the other without establishing a link. As a result, no link must be established before transmitting data from the sender to the recipient. It is not a dependable network service since it does not guarantee the delivery of data packets to the receiver, and data packets will arrive at the receiver in any order. As a result, the data packet does not follow a predetermined direction. The transmitted data packet is not received by the recipient in a connectionless operation due to network interference, and the data may be lost.

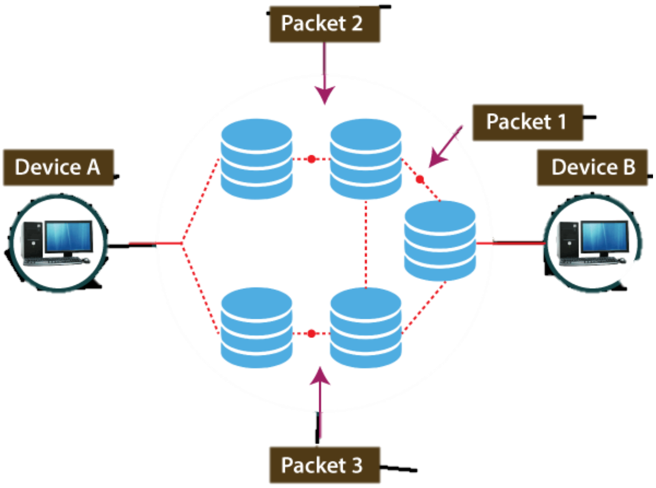

Figure 8 Connectionless Communication

Since it is a connectionless service, a sender can transmit some data to the receiver without first forming a link. The sender's data would be included in the packet or data sources that hold the receiver's address. Data can be sent and received in any order in a connectionless operation. However, it does not promise that the packets will be sent to the correct location.

Example of connectionless service is : UDP

The UDP (User Datagram Protocol) is a connectionless protocol that allows two or more devices to communicate without having to create a link. A sender sends data packets to the receiver with the destination address in this protocol. A UDP does not guarantee that data packets are sent to the right destination or that the sender's data is acknowledged.

The difference between connection-oriented and connectionless is shown in Table 1.

Table 1 Connection-oriented vs Connectionless service

Connection-oriented service	Connectionless Service
It is built on the telephone system in terms of architecture and development.	It is a postal system-based operation.
Until sending data over the same or a separate network, it is used to provide an end-to-end link between the senders and the receiver.	It is used to transmit data packets from senders to receivers without establishing a link.
It establishes a simulated connection between the sender and the recipient.	Between the sender and the recipient, no virtual link or route is established.
Before sending data packets to the recipient, it needs authentication.	Before transmitting data packets, it does not require authentication.
The sender's data packets are returned in the same order that they were delivered.	The order in which data packets are received differs from the order in which they are sent by the sender.
The data packets must be sent over a larger bandwidth.	The data packets must be sent over a low-bandwidth connection.
As it ensures data packets pass from one end to the other with a link, it is a more secure connection service.	Since it does not guarantee the transmission of data packets from one end to the other in order to create a link, it is not a secure connection service.

Since it provides an end-to-end link between the sender and receiver through data transmission, there is no interference.	Owing to the lack of an end-to-end link between the source and receiver for data packet transmission, there could be congestion.
A connection-oriented service is the Transmission Control Protocol (TCP).	Connectionless services include the User Datagram Protocol (UDP), Internet Protocol (IP), and Internet Control Message Protocol (ICMP).

10.4 Reliable vs Unreliable Delivery

- If the application layer program needs reliability, we use a reliable transport layer protocol by implementing flow and error control at the transport layer.
- This means a slower and more complex service.
- In the Internet, there are three common different transport layer protocols. UDP is connectionless and unreliable.
- TCP and SCTP are connection oriented and reliable.

10.5 TCP/IP protocol suite

The original TCP IP protocol suite defines two transport layer protocols. One is UDP, and the other is TCP, which we discussed when I taught you about the OSI model.

Position of UDP in the TCP/IP protocol suite

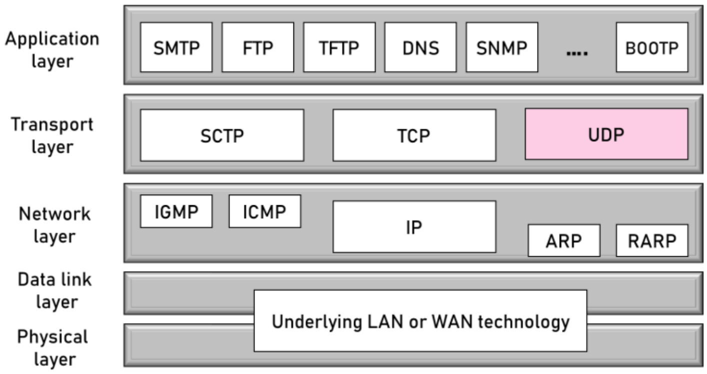

Figure 9 Position of UDP in the TCP/IP protocol suite

In this case, we'll start with a UDP, which is the easier of the two. Before we get into the TCP. We've also developed a new transport protocol called SCTP.

UDP

So, first and foremost, we'll talk about the UDP protocol, which stands for user datagram protocol in its full form. A connectionless and insecure transport protocol is what it's called. It adds little to the IP's services other than providing process-to-process communication rather than HOST-to-HOST communication. It also does only rudimentary error checking. Why does a mechanism want to use UDP if it is too powerless? As a result of UDP. If there is a downside. It comes with a slew of benefits. The main benefits are that UDP is a very basic protocol that uses very little overhead.

If a process needs to send a small message and isn't concerned about acknowledgement or reliability, it should use UDP. Sending a small message with UDP requires much less contact between the sender and the recipient than sending a small message with TCP or SCTP. We have well-known ports for UDP, as seen in the Figure 10.

Few ports may be used for both UDP and TCP. For example, FTP can use port 21 for either UDP or TCP. SNMP uses two port numbers, 161 and 162, with different purposes.

Port	Protocol	Description
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
20	FTP, Data	File Transfer Protocol (data connection)
21	FTP, Control	File Transfer Protocol (control connection)
23	TELNET	Terminal Network
25	SMTP	Simple Mail Transfer Protocol
53	DNS	Domain Name Server
67	BOOTP	Bootstrap Protocol
79	Finger	Finger
80	HTTP	Hypertext Transfer Protocol
111	RPC	Remote Procedure Call

Figure 10 Well Known Ports for UDP

UDP Header

The UDP header is an 8-byte defined and basic header, while the TCP header will range from 20 to 60 bytes. The first 8 bytes contain all required header information, while the remainder is text. Since each UDP port number field is 16 bits long, the range for port numbers is 0 to 65535; port number 0 is reserved. Port numbers are used to differentiate between various user queries or processes.

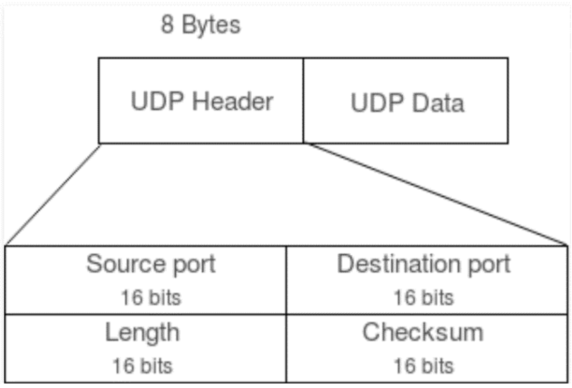

Figure 11 UDP Datagram Protocol

Source Port : Source Port is a two-byte field that identifies the source's port number.

Destination Port: This is a two-byte field that identifies the destined packet's path.

Length: The length of UDP, including the header and data, is measured in bytes. It's a region of 16 bits.

Note: Checksum calculation is not needed in UDP, unlike TCP. UDP does not have error management or flow control. As a result, UDP relies on IP and ICMP to record errors.

10.6 Applications of UDP

- When the size of the data is small, there is less concern for flow and error management, because it is used for basic request-response communication.
- Since UDP supports packet swapping, it is a good protocol for multicasting.
- Few routing upgrade protocols, such as RIP, use UDP (Routing Information Protocol).
- Often used in real-time systems where uneven delays between portions of a sent message cannot be tolerated.
- Following implementations uses UDP as a transport layer protocol:

- o NTP (Network Time Protocol)
- o DNS (Domain Name Service)
- o BOOTP, DHCP.
- o NNP (Network News Protocol)
- o Quote of the day protocol
- o TFTP, RTSP, RIP.

Some activities may be performed by the application layer using UDP, such as route tracing, route recording, and route time stamping.

UDP receives datagrams from the Network Layer, attaches their headers, and sends them to the customer. As a result, it works quickly.

If you delete the checksum field from UDP, it becomes a null protocol.

1. Reduce the amount of computing resources used.
2. When transferring via Multicast or Broadcast.
3. Real-time packet transfer, primarily in multimedia applications

10.7 TCP 3-Way Handshake Process

TCP stands for Transmission Control Protocol, which means it does everything to ensure the data is transmitted in a secure manner.

The latest TCP/IP suite paradigm governs the mechanism of communication between devices over the internet (stripped out version of OSI reference model).

The Application layer is the top layer of the TCP/IP architecture, from which client-side network applications such as web browsers create connections with the server. The information is moved from the application layer to the transport layer, which is where our subject appears.

TCP and UDP (User Datagram Protocol) are two important protocols in this layer, with TCP being the most common (since it provides reliability for the connection established). However, UDP can be used to test the DNS registry in order to get the binary equivalent of the website's domain name.

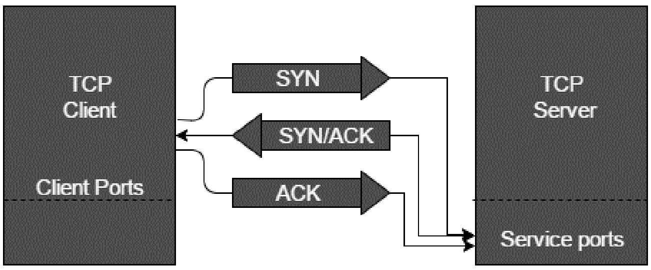

Figure 12 TCP Handshake

Positive Acknowledgement with Re-transmission is a feature of TCP that ensures secure communication (PAR). The transport layer's Protocol Data Unit (PDU) is known as segment. Still, once it gets an acknowledgment, a system using PAR can resend the data unit. If the data unit received at the receiver's end is corrupted (it scans the data with the transport layer's checksum feature for Error Detection), the section is discarded. As a result, the sender must resend the data device about which there is no positive acknowledgment.

As you can see from the above mechanism, three segments are exchanged between the sender (client) and receiver (server) in order to create a secure TCP link as seen in Figure 13.

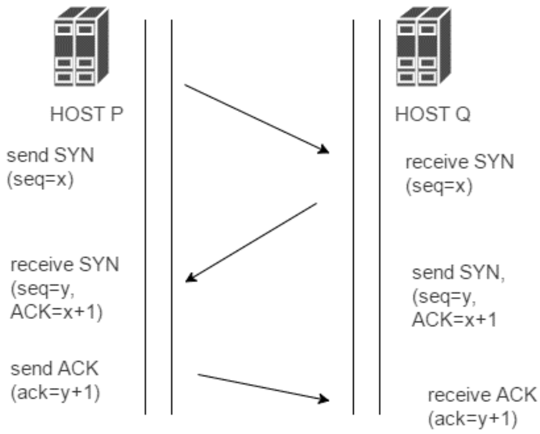

Figure 13 Working

Stage 1 (SYN): In the first step, the client needs to create a link with the server, so it sends a segment with SYN (Synchronize Sequence Number), which tells the server that the client is likely to begin contact and with what sequence number it will begin segments.

Step 2 (SYN + ACK): When the server receives a client message, it sets the SYN-ACK signal bits. Acknowledgement (ACK) denotes the answer of the segment it got, while SYN denotes the sequence number from which it is likely to begin the segments.

Step 3 (ACK): In the final step, the client accepts the server's response and the two of them create a secure link to begin the data transfer. The relation parameter (sequence number) for one direction is established and acknowledged in steps 1 and 2. The relation parameter (sequence number) for the other direction is established and acknowledged in steps 2 and 3. A full-duplex connectivity is developed with these.

Note

When forming relations between the client and the server, the initial sequence numbers are chosen at random.

Summary

- The transport layer of the OSI reference model allows direct data transmission between source and destination machines by using network layer services such as IP to pass PDUs of data between the two communicating machines.
- The transport layer is an end-to-end or source-to-destination layer. To ensure full data sharing, the OSI Transport layer protocol (ISO-TP) manages end-to-end monitoring and error checking. It allows “peer to peer” contact with the destination machine's transport entity (remote peer).
- The transport layer adds a secure layer on top of the network layer's insecure networks. Option negotiation among various quality of service criteria provides consumer applications with efficient, dependable, and cost-effective transportation services.
- Flow control regulates data transfer between devices, ensuring that the transmitting device sends no more data than the receiving device can handle. Data from multiple applications can be sent over a single physical connection using multiplexing.
- On top of the network layer, transport primitives are an efficient means of transmitting data. The transport layer facilities tend to be identical to those delivered at the data link layer. However, they vary in several respects, with the data link layer using physical channels to bind two routers and the transport layer using subnets.

- UDP is a connectionless, insecure protocol that helps processes run faster by reducing the burden on their CPUs. The performance challenges, which lack a scientific paradigm to back them up, are backed up by personal perceptions and examples. They try to solve problems in computer networks by testing network efficiency, designing systems for improved performance, handling TPDUs quickly, and developing protocols for future high-performance networks.

Keywords

Addressing: Addressing or tagging a frame is handled by the Transport Layer.

Connection Establishment Delay: That's the length of time it takes for the destination system to accept that a connection has been demanded. Obviously, the shorter the wait time, the better the service.

Connection Establishment Failure Probability: Because of network congestion, a lack of table space, or other internal issues, the link does not develop within the defined delay.

Connection Establishment/Release: A naming function is used in the transport layer for forming and releasing links across the network, so that a process on one computer may indicate with whom it wants to interact.

Demultiplexing: Where multiple links are multiplexed, demultiplexing is needed at the receiving end.

Error Control: To prevent errors caused by missing or overlapping segments, the transport layer assigns specific segment sequence numbers to individual message packets, forming virtual circuits with only one virtual circuit per session.

Flow Control: The fundamental principle of flow control is to keep a quick and slow mechanism in synergy. The transport layer makes it possible for a fast process to keep up with a sluggish one.

Fragmentation: When the transport layer receives a large message from the session layer, it divides the message into smaller units as required.

Multiplexing: The transport layer provides several network links to increase throughput.

Throughput: In a given time period, it specifies the amount of bytes of user data transmitted per second. It is calculated separately for each contact source.

Transmission Control Protocol: It allows a secure data distribution service with error detection and correction from beginning to end.

User Datagram Protocol(UDP): It is an insecure connectionless datagram protocol in which the transmitting terminal does not validate if data has been received by the receiving terminal.

Review Questions

1. When the facilities delivered at both layers are almost identical, how is the transport layer different from the data link layer?
2. Why transport layer is required when both the network and transport layers provide connectionless and connection oriented services?
3. What are the different quality of services parameters at the transport layer?
4. Why UDP is used when it provides unreliable connectionless service to the transport layer?
5. What is the purpose of flow control?
6. Describe the TCP and its major advantages over UDP.

Self-Assessment

1. UDP known as
    a. User Datagram Protocol
    b. Unity Data Packet

c. User datagram Packet
d. None of Above

2. TCP is a
   a. Telnet Control Protocol
   b. Transmission Cent Protocol
   c. Transmission Control Protocol
   d. None of above
3. FTP uses port
   a. 22
   b. 23
   c. 24
   d. 21
4. Which of the following is false with respect to UDP?
   a. Connection-oriented
   b. Unreliable
   c. Transport layer protocol
   d. Low overhead
5. What is the main advantage of UDP?
   a. More overload
   b. Reliable
   c. Low overhead
   d. Fast
6. The ______ field is used to detect errors over the entire user datagram.
   a. udp header
   b. checksum
   c. source port
   d. destination port
7. Which is the correct expression for the length of UDP datagram?
   a. UDP length = IP length – IP header's length
   b. UDP length = UDP length – UDP header's length
   c. UDP length = IP length + IP header's length
   d. UDP length = UDP length + UDP header's length
8. Port number used by Network Time Protocol (NTP) with UDP is ________
   a. 161
   b. 123
   c. 162
   d. 124
9. What is the header size of a UDP packet?
   a. 8 bytes
   b. 8 bits
   c. 16 bytes
   d. 124 bytes
10. Beyond IP, UDP provides additional services such as _______
    a. Routing and switching
    b. Sending and receiving of packets
    c. Multiplexing and demultiplexing

d. Demultiplexing and error checking

11. Beyond IP, UDP provides additional services such as _______
    a. Routing and switching
    b. Sending and receiving of packets
    c. Multiplexing and demultiplexing
    d. Demultiplexing and error checking
12. IP Security operates in which layer of the OSI model?
    a. Network
    b. Transport
    c. Application
    d. Physical

Self-Assessment Answers

1.	a	2.	c	3.	d	4.	a
5.	c	6.	b	7.	a	8.	b
9.	a	10.	d	11	d	12.	a

Further Readings

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